

# SILICON DREAMS

## Study Guide

MODULE 2 · WEEK 2 · CM-HW-101-M2

SG-M2-08 pairs with TB-M2-08

## M2 Escape Checklist & Reflection

Merge fixes, push, verify the regression matrix, file the post-mortem.

### Learning objectives

- Integrate all floor fixes onto a single branch with a clean history.
- Run the full M2 regression matrix locally before pushing.
- Re-run DRC against the patched RTL.
- File a grading-ready post-mortem in notes/escape-log.md.

### Final integration branch

```
git checkout -b m2-escape
# If you worked on per-floor branches, merge them here
for f in m2-floor-b2 m2-floor-5 m2-floor-1 m2-silent; do
  git show-ref --verify --quiet refs/heads/$f && git merge $f
done

# Run the full matrix locally
cd test
make clean
make smoke
make fault
make yosys-report

# If everything is green, push
git push origin m2-escape
```

### Pre-push regression checklist

- ☐ src/elevator.v: reset polarity fixed (Floor B2).
- ☐ src/elevator.v: state register re-encoded to one-hot with illegal-state trap (Floor 5).
- ☐ src/elevator.v: requested\_floor validity gate added, uio\_out[7] surfaces error (Floor 1).
- ☐ src/elevator.v: delay register narrowed (Silent Floor — optional).
- ☐ test/fault\_injection/test\_floor\_b2.py: present and PASSes.
- ☐ test/fault\_injection/test\_floor\_5.py: present and PASSes.
- ☐ test/fault\_injection/test\_floor\_1.py: present and PASSes.
- ☐ test/fault\_injection/test\_silent.py (optional): present, attests Yosys delta.
- ☐ notes/escape-log.md: one section per floor with symptom / root cause / fix / proof / time.
- ☐ notes/yosys-pre.txt + notes/yosys-post.txt: both committed if Silent Floor attempted.
- ☐ harness/ (bonus): new fault class + README + test\_composite.py.
- ☐ make smoke: still PASSes. No regression from any of the above.

DRC re-run

```
cd test
make hardening
# Or push a PR labelled run-drc and let CI do the work
# Expected:

CHECK 1: State Encoding (FLOOR 5)
- Count DFFs in state machine (should be 4 for one-hot)

CHECK 2: One-Hot Parameters in RTL
- Verify one-hot values exist (4'b0001, 4'b0010, etc.)

CHECK 3: State Register Width
- Check that state is reg [3:0] (4 bits)

CHECK 4: Illegal State Trap
- Check for default: next_state = IDLE_STATE;

CHECK 5: Input Validity Gate
- Verify requested_floor_valid exists

CHECK 6: Reset Polarity
- Check for negedge rst_n (active-low reset)
```

CHECK 7: Resource Utilization

- Count total DFFs (should be < 50)

If it still fails, file a tick in the course Discord under #m2-drc with the failing log.

Rubric — what the graders check

|                       |                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| Minimum pass (600 XP) | Three floors cleared (B2+5+1). Three fault tests green. Smoke test green. Escape log covers all three.          |
| Distinction (900 XP)  | All four floors cleared including Silent Floor. Yosys pre/post attached. Clean DRC. Escape log covers all four. |
| Bonus (1000 XP)       | Distinction plus harness/ package with a new fault type and test_composite.py.                                  |
| DRC clean first try   | +100 XP. Same habit as Module 1 bonus. Pay attention to utilisation before pushing.                             |

Reflection prompts (include in escape log)

1. Which floor was hardest for you, and why? (One paragraph. Be specific about the confusion — graders learn from your struggle.)
2. If you had to teach the Floor 5 re-encoding to a teammate in five minutes, what is the one-line takeaway and the one figure you would draw?
3. What would you do differently in a future Module-1-style design to avoid introducing the Floor 1 bug in the first place? (Hint: where does the validation live?)
4. How much of the Module 2 harness do you think you will carry into Module 3? (Honest answer; no wrong answers.)

The last thing to do

Push the branch. Watch the badge. If it is green, you have escaped Module 2. AXIOM will see you in Module 3. The chapter TB-M2-08 closes with AXIOM's last line — re-read it once the badge turns green. It reads differently then.